

- 2 Melanism (dark-coloured wings) in a species of moth is controlled by a gene with two alleles. The dominant allele codes for the melanic form and the recessive allele codes for the non-melanic form (pale, speckled wings). Birds are a predator of this species.

The melanic form is better camouflaged in areas polluted by carbon deposits from smoke. The non-melanic form is better camouflaged in areas unpolluted by smoke.

The frequencies of these two forms of moth were investigated in two populations living in different areas, **X** and **Y**, by using moth traps and counting the numbers of each type of moth. The investigation was carried out over a period of time corresponding to twenty-four generations of this species.

(a) Suggest **two** variables in this investigation that should be standardised.

.....
.....
.....[2]

During this time period, human activity increased in both areas **X** and **Y** as towns and roads were developed. During generation 1, a large factory started releasing lots of smoke in area **X**.

Fig. 2.1 shows the results of the investigation.

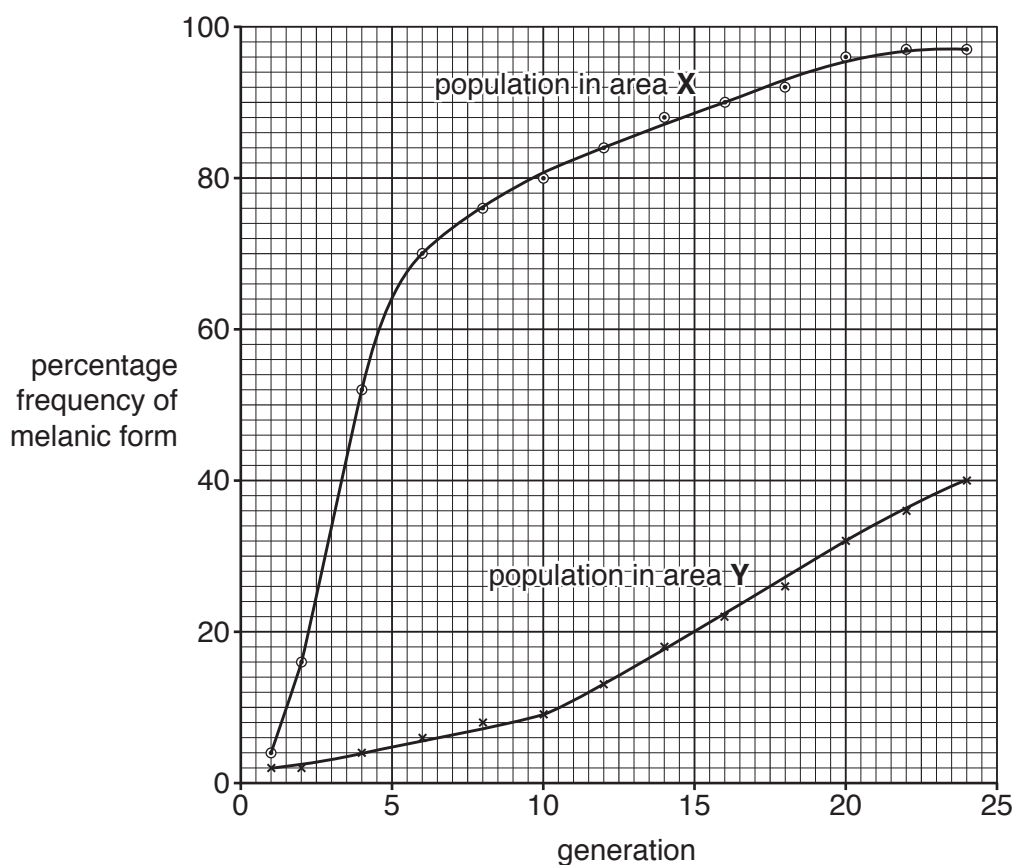


Fig. 2.1

(b) Describe **and** explain the changes in the **two** populations, as shown in Fig. 2.1.

.....[3]

(c) If the factory in area X closed during generation 10, sketch a line on Fig. 2.1 to predict the likely effect on the frequency of the melanic form of the moth in area X for the next 14 generations. [1]

(d) The investigators assumed that the changes in the frequencies of the different types of moth in both areas **X** and **Y** were the result of predation by birds.

Suggest two other possible reasons for the increase in frequency of the melanic form.

1

.....

.....

2

.....

.....

[2]

- (e) The investigators carried out test crosses by breeding a heterozygous male melanic moth with several female non-melanic moths. The number of moths of each form in the next generation was recorded.

melanic 56
non-melanic 48

- (i) Complete Table 2.1 to calculate the chi-squared (χ^2) value for determining whether or not these results were significantly different from those expected for crosses of this type.

The formula for calculating χ^2 is:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

ν = degrees of freedom c = number of classes

Table 2.1

category	O	E	$\frac{(O - E)^2}{E}$
melanic	56		
non-melanic	48		
$\chi^2 =$			

[3]

- (ii) Table 2.2 shows some critical values for χ^2 at different probabilities.

Table 2.2

degrees of freedom	probability						
	0.99	0.95	0.90	0.10	0.05	0.01	0.001
1	0.0002	0.0039	0.0158	2.706	3.841	6.635	10.827
2	0.0201	0.1026	0.2107	4.605	5.991	9.210	13.815

Use Table 2.2 to explain whether or not the χ^2 value calculated in Table 2.1 is significant.

.....

[1]

[Total: 12]

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